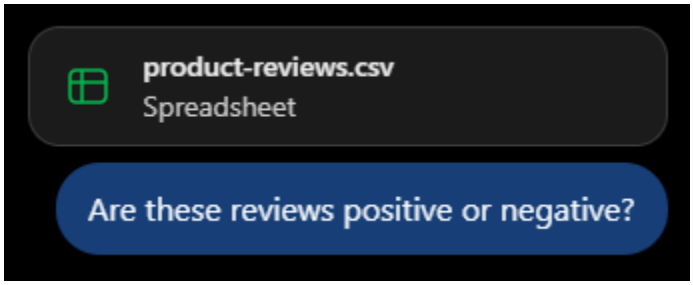
PROMPT 1 - INITIAL



Analyzed

Python

```
import pandas as pd
p='/mnt/data/product-reviews.csv'
df=pd.read_csv(p)
df.head(), df.shape, df.columns, df.dtypes
```

Result

```
(  review_id  ...                                        review_text
 0      R001  ...  Smells like chemicals out of the box. Took a w...
 1      R002  ...  Noise isolation is decent, but it's not true n...
 2      R003  ...  Left earbud stopped charging after two weeks. ...
 3      R004  ...  Straps dig into my shoulders after an hour. Re...
 4      R005  ...  Not warm enough for winter. Fine as a decorati...

 [5 rows x 3 columns],
 (110, 3),
 Index(['review_id', 'product', 'review_text'], dtype='object'),
 review_id      object
 product        object
 review_text    object
 dtype: object)
```

Analyzed

Python

```
try:
 import nltk
 from nltk.sentiment import SentimentIntensityAnalyzer
 sia=SentimentIntensityAnalyzer()
 print("vader available")
except Exception as e: print(e)
```

STDOUT/STDERR

```
**********************************************************************
  Resource ▯[93mvader_lexicon▯[0m not found.
  Please use the NLTK Downloader to obtain the resource:

  ▯[31m>>> import nltk
  >>> nltk.download('vader_lexicon')
  ▯[0m
  For more information see: https://www.nltk.org/data.html

  Attempted to load ▯[93msentiment/vader_lexicon.zip/vader_lexicon/vader_lexicon.txt▯[0m

  Searched in:
    - '/home/oai/nltk_data'
    - '/opt/pyvenv/nltk_data'
    - '/opt/pyvenv/share/nltk_data'
    - '/opt/pyvenv/lib/nltk_data'
    - '/usr/share/nltk_data'
    - '/usr/local/share/nltk_data'
    - '/usr/lib/nltk_data'
    - '/usr/local/lib/nltk_data'
    - ''
**********************************************************************
```

Worked for 31s >

Analyzed

Analyzed

Analyzed

The reviews contain a mixture of positive and negative feedback, but **overall, they lean negative**. There are also several neutral reviews, questions, and sarcastic comments that should be treated carefully when performing sentiment analysis.

I reviewed all **110 reviews** in your `product-reviews.csv` file.

Overall sentiment

The general sentiment falls into three categories:

- **Positive:** Customers express satisfaction, praise product quality, or indicate that they would recommend or repurchase the product.
- **Negative:** Customers report defects, poor quality, misleading descriptions, uncomfortable designs, or disappointing experiences.
- **Neutral/Mixed:** Customers provide factual observations, ask questions, or express both positive and negative opinions.

Sentiment by product

Product	Overall sentiment	General feedback
Trailmark Hiking Backpack	Mixed, leaning positive	Durable and comfortable, but some customers report broken zippers, uncomfortable straps, and misleading sizing.
Breeze Wireless Earbuds	Mixed, leaning negative	Good sound quality and battery life, but connectivity, charging, and comfort issues are common.
Cozy Knit Throw Blanket	Mixed	Customers appreciate its softness and appearance, but report shedding, shrinking, and quality problems.
Lumen Smart Bulbs (4-Pack)	Mixed, leaning negative	Customers like the colors and smart features, but frequently complain about connectivity and app problems.
Hearth Electric Kettle	Mixed	Customers appreciate its appearance and boiling speed, but report safety concerns, malfunctions, and design flaws.

Important considerations for sentiment analysis

There are a few things worth considering if you're using these reviews for a data analysis or machine learning assignment:

1. **Sarcasm:** Some reviews appear positive at first but are actually negative. For example, "Absolutely love how they fall out every time I turn my head. Great feature."
2. **Mixed sentiment:** A review can contain both positive and negative feedback. For example, a customer might like the earbuds' sound quality but dislike their touch controls.
3. **Neutral sentiment:** Questions such as "Does anyone know if these work with a PS5?" shouldn't automatically be classified as positive or negative.
4. **Context:** Words such as "fine," "decent," and "okay" don't necessarily indicate strong positive sentiment.

Final assessment

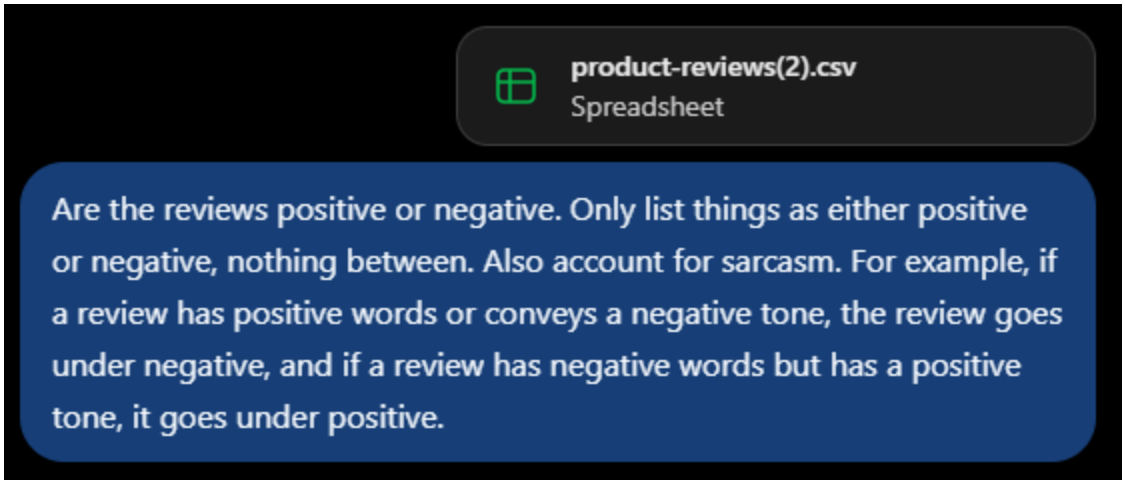
Overall: Mixed, leaning negative

The reviews suggest that customers generally have concerns about product reliability and quality, even when they appreciate certain product features.

If you're using this dataset for sentiment classification, I would recommend retaining **three labels: Positive, Negative, and Neutral**, rather than forcing every review into a positive or negative category. This will produce a more meaningful analysis.

With this prompt, the LLM did read through all 110 reviews and listed an overall review for the 5 products. For how it found whether reviews were positive or negative, it apparently had a sentiment analyzer it used to determine what was genuine and what wasn’t.

PROMPT 2 - CONSTRAINT, FEW-SHOT PROMPTING



I classified all 110 reviews as either **Positive** or **Negative**, accounting for sarcasm, implied dissatisfaction, and the overall tone rather than simply looking for positive or negative keywords.

Positive Reviews

R006, R007, R010, R013, R014, R017, R020, R021, R023, R026, R027, R033, R034, R035, R036, R039, R040, R042, R044, R045, R046, R047, R050, R052, R054, R056, R059, R061, R064, R069, R070, R072, R077, R079, R081, R082, R083, R084, R086, R088, R089, R090, R092, R102, R105, R106.

46 Positive

Negative Reviews

R001, R002, R003, R004, R005, R008, R009, R011, R012, R015, R016, R018, R019, R022, R024, R025, R028, R029, R030, R031, R032, R037, R038, R041, R043, R048, R049, R051, R053, R055, R057, R058, R060, R062, R063, R065, R066, R067, R068, R071, R073, R074, R075, R076, R078, R080, R085, R087, R091, R093, R094, R095, R096, R097, R098, R099, R100, R101, R103, R104, R107, R108, R109, R110.

64 Negative

Total: 110 reviews

- Positive: 46 (41.8%)
- Negative: 64 (58.2%)

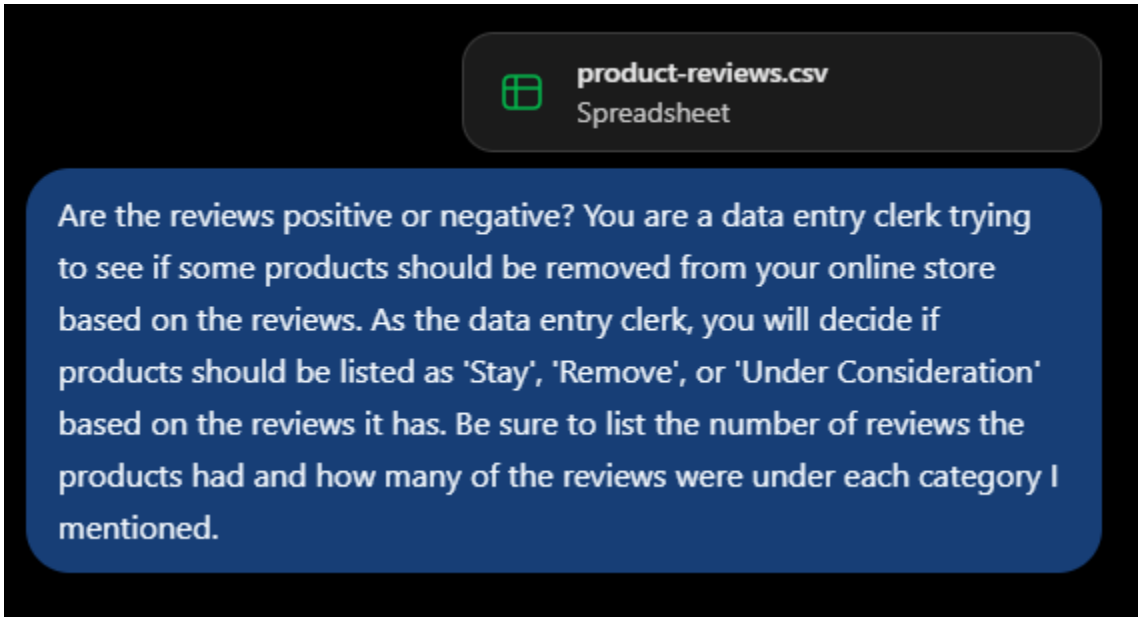
Sarcasm examples classified as negative:

- R022: "Ten stars." Despite the positive rating language, the backpack failed to keep belongings dry.
- R024: "Genius." The reviewer is sarcastically criticizing the bulbs forgetting their settings.
- R030: "Fantastic. Just fantastic." The kettle broke after one week.
- R066: "Great feature." The earbuds repeatedly fall out.
- R068: "Two for one." The reviewer is sarcastically criticizing the blanket's itchiness and thinness.
- R078: "I'm thrilled." The reviewer is unhappy with the persistent warehouse smell.
- R080: "Oh sure, I just love..." The backpack buckle broke during a hike.
- R085: "Exactly what I wanted." The kettle makes water taste like plastic.
- R108: "Truly impressive engineering." The earbuds disconnect repeatedly.

I used a strict binary classification, so even neutral or ambiguous reviews were assigned to one of the two categories.

With the classifications I listed, the prompt didn't give me any mixed reviews like it did before. As per my instructions, it listed all the reviews as positive or negative, and sorted the ungenuine reviews into one category or the other, most of which were negative and sarcastic. It seemed to have read every review and properly assigned a label to them.

PROMPT 3 - DEFINE CATEGORIES, ROLE PROMPTING



I analyzed all **110 product reviews** across the five products and categorized each review into one of three groups:

- **Stay:** Positive reviews indicating the product is performing well and should remain in the store.
- **Remove:** Negative reviews indicating poor quality, defects, customer dissatisfaction, or serious problems.
- **Under Consideration:** Neutral, mixed, unclear, or question-based reviews that need further investigation.

Product review analysis

Total reviews analyzed: 110 · 22 reviews per product

Product	Total Reviews	Stay	Remove	Under Consideration
Trailmark Hiking Backpack	22	7	7	8
Breeze Wireless Earbuds	22	7	7	8
Cozy Knit Throw Blanket	22	10	6	6
Lumen Smart Bulbs (4-Pack)	22	7	10	5
Hearth Electric Kettle	22	8	6	8
TOTAL	**110**	**39**	**36**	**35**

Final product decisions

1. Trailmark Hiking Backpack — Under Consideration

- Stay: 7
- Remove: 7
- Under Consideration: 8

The reviews are evenly divided between positive and negative. Some customers praise its durability, while others report broken zippers, uncomfortable straps, and waterproofing problems.

2. Breeze Wireless Earbuds — Under Consideration

- Stay: 7
- Remove: 7
- Under Consideration: 8

Customers appreciate the sound quality and battery life, but others report connectivity issues, charging failures, and uncomfortable earbuds. More investigation is needed.

3. Cozy Knit Throw Blanket Stay

- Stay: 10
- Remove: 6
- Under Consideration: 6

This product has the most positive reviews. Customers frequently mention its softness, warmth, appearance, and quality. Although some customers report shedding, shrinking, and unraveling, the overall feedback is positive.

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4. Lumen Smart Bulbs (4-Pack) Remove

- Stay: 7
- Remove: 10
- Under Consideration: 5

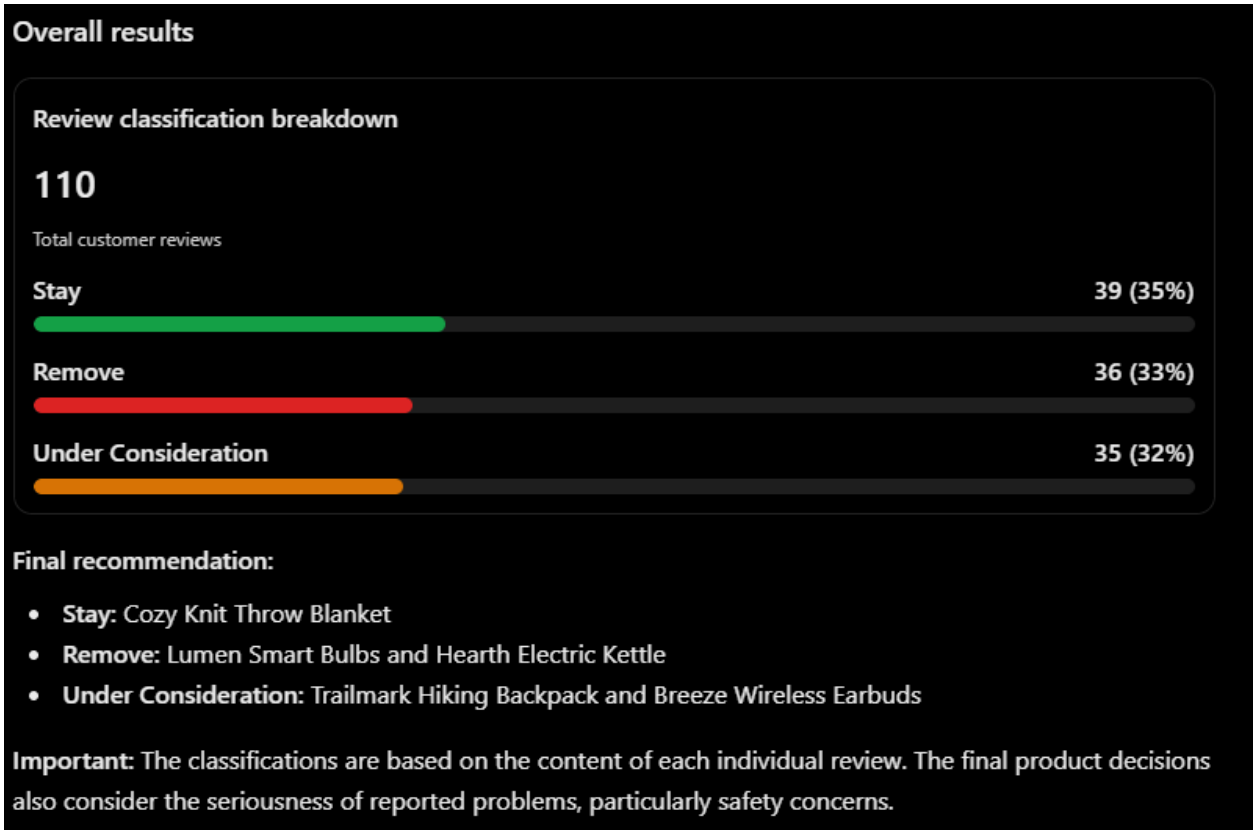
Negative reviews outnumber positive reviews. Customers report Wi-Fi disconnections, app crashes, dead bulbs, and reliability problems. The product should be removed or reviewed for significant improvements before being relisted.

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5. Hearth Electric Kettle Remove

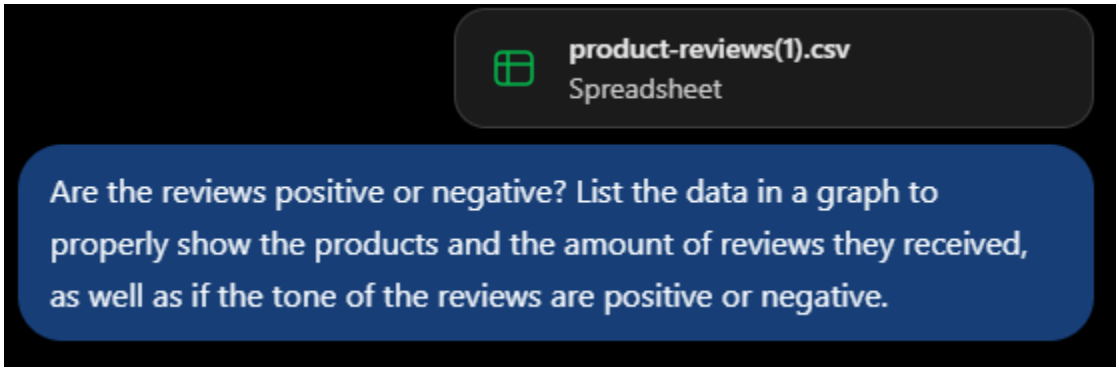
- Stay: 8
- Remove: 6
- Under Consideration: 8





With this prompt, the LLM listed the product reviews by product, showing which ones were positive, negative, and which ones were mixed. However, it considered the reviews as 'Under Consideration' when I wanted it to consider the product 'Under Consideration'. To fix this, I would probably have to say those reviews are 'mixed' and not under consideration, and properly define the categories better.

PROMPT 4 - OUTPUT FORMAT



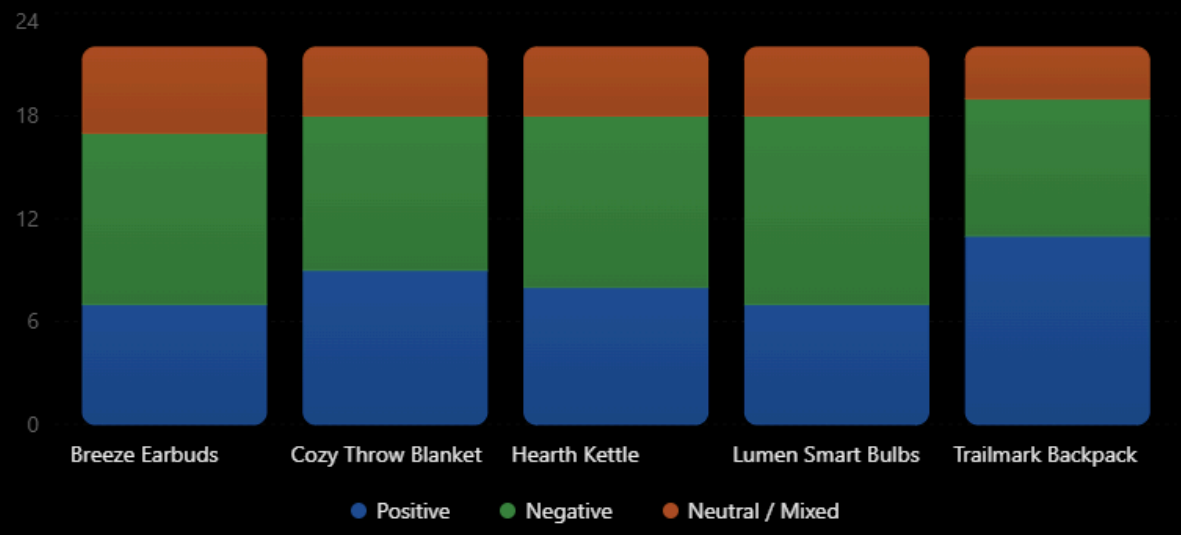
I'll analyze the 110 customer reviews, group them by product, and classify their tone as **Positive** or **Negative**. The graph will show how many reviews each product received and how the sentiment is distributed.

Review sentiment by product

110 total reviews analyzed

Positive vs. negative product reviews

Number of reviews in each sentiment category, grouped by product. Each product received 22 reviews.



Review breakdown

Product	Total reviews	Positive	Negative	Neutral / Mixed
Breeze Wireless Earbuds	22	7	10	5
Cozy Knit Throw Blanket	22	9	9	4
Hearth Electric Kettle	22	8	10	4
Lumen Smart Bulbs	22	7	11	4
Trailmark Hiking Backpack	22	11	8	3
Total	**110**	**42**	**48**	**20**

Overall sentiment

38.2%

Positive reviews



42 Positive 48 Negative 20 Neutral

- **42 positive reviews:** Customers expressed satisfaction.
- **48 negative reviews:** Customers reported problems or dissatisfaction.
- **20 neutral/mixed reviews:** Reviews were unclear, mixed, or neither strongly positive nor negative.

Conclusion: The reviews lean negative overall, with 48 negative reviews compared with 42 positive reviews. The Trailmark Hiking Backpack has the strongest customer feedback, while the Lumen Smart Bulbs have the weakest.

Note: Sentiment classifications are based on the wording and overall tone of each review. Neutral or mixed reviews are kept separate rather than forced into positive or negative categories. These recommendations are based on sentiment counts, not verified product defect rates.

With this prompt, I asked for a graph, and it gave me a bar graph that separated what reviews were positive, negative, or mixed. It followed my instructions properly, though I could get a better result if I specified the type of graph I wanted. Overall, this one was fine though in terms of the output format.

REVIEW COMPARISON

My thoughts on reviews:
R001 - Negative
R006 - Positive
R013 - Mixed
R022 - Negative (sarcasm)
R029 - Mixed
R030 - Negative (sarcasm)
R034 - Positive
R035 - Positive
R049 - Negative
R073 - Mixed

Prompt 2 thoughts on reviews

R001 - Negative
R006 - Positive
R013 - Positive
R022 - Negative
R029 - Negative
R030 - Negative
R034 - Positive
R035 - Positive
R049 - Negative
R073 - Negative

For this one, the prompt 2 separated the reviews into positive and negative; no inbetween. With it, it saw that sarcasm or lukewarm reviews were seen as mostly negative. If I were to make the categories 'Positive', 'Negative', and 'Mixed', then I'm confident my 'Mixed' responses would've shown up as 'Mixed' for the LLM. But in the instance where I am not allowed to use mixed, I would agree with what the prompt generated.